Who comes back within 30 days? Prior admissions, length of stay and diabetes medication in 69,290 hospitalised diabetic patients

Group name	[Group name]
Dataset	UCI Machine Learning Repository, "Diabetes 130-US Hospitals for Years 1999–2008": archive.ics.uci.edu/dataset/296
Code	[github.com/.../cse558-midterm]: notebook `pre_fin_final_with_processed_hypothesis_tests.ipynb` and `README.md`

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1. Dataset

Source: UCI Machine Learning Repository, dataset 296 (Strack et al., *BioMed Research International*, 2014). Ten years (1999–2008) of inpatient encounters of diabetic patients from 130 US hospitals, extracted from the Health Facts clinical database. One row is one hospital *encounter* lasting 1–14 days in which diabetes was recorded and a laboratory test and medication were given. The response, `readmitted`, records whether the patient came back within 30 days (`<30`), after more than 30 days (`>30`) or not at all (`NO`).

What is raw about it: missing values are stored as `"?"`; three ID columns use codes such as "Not Available", "NULL" or "Not Mapped" that are missing data in disguise; the same patient appears up to 40 times; patients who died are labelled `NO`; diagnoses are raw ICD-9 codes (717–790 levels per column); 23 medication columns are text (`No/Steady/Up/Down`).

	Raw file	After encoding
Instances (rows)	101,766 encounters	69,290 patients
Attributes (columns)	50	128 (125 features + 3 target)
Instances $\times$ attributes	5,088,300	8,869,120

Size check: $69{,}290 \times 128 = 8{,}869{,}120 \geq 3{,}000{,}000$. Even the 125 input features alone give $69{,}290 \times 125 = 8{,}661{,}250$, so the requirement is met with room to spare. One-hot encoding of 18 nominal attributes into 110 indicator columns is what takes the width from 35 to 128.

Key features

Feature	Type	Real-world meaning
`readmitted`	nominal	`NO` / `>30` / `<30` days to next admission; our response
`number_inpatient`	discrete	inpatient admissions in the year before the encounter
`number_emergency`, `number_outpatient`	discrete	emergency and outpatient visits in the previous year
`time_in_hospital`	discrete	length of stay, 1–14 days
`num_lab_procedures`, `num_medications`	discrete	lab tests and distinct drugs during the stay
`age`	ordinal	10-year bands, `[0-10)` to `[90-100)`
`insulin`	nominal	insulin prescribed and its dose change: `No`, `Steady`, `Up`, `Down`
`diabetesMed`, `change`	binary	any diabetes drug prescribed; any change in diabetes drugs
`A1Cresult`	nominal	HbA1c result, or `None` if not measured
`diag_1`–`diag_3`	nominal	primary and secondary ICD-9 diagnoses

2. Data preparation

Steps run in the order listed. Every count is from the notebook run on the raw file.

Issue found	What we did	Why, and not the alternative
Exact duplicate rows?	compared all columns except `encounter_id`; 0 found	`encounter_id` is unique per row, so a check that includes it can never find a duplicate.
2,423 encounters ended in death or hospice (discharge codes 11, 13, 14, 19, 20, 21)	removed	these patients *cannot* be readmitted; 2,337 of them are labelled `NO`, which would inflate the "not readmitted" group with people who died (same exclusion as Strack et al.).
71,518 patients for 101,766 encounters (up to 40 each)	kept each patient's first encounter (lowest `encounter_id`); 29,353 rows removed	repeat visits of one patient are not independent, and every test in Section 5 assumes independent observations. Frequent attenders would also be over-weighted.
`weight` 96.0% missing	column dropped	nothing credible can be imputed from 4% observed values.
13 drug columns with $\geq$99.5% `No`	3 constant ones dropped; the other 10 tested against the outcome (χ^2, Monte Carlo when sparse, Benjamini–Hochberg); only `acarbose` kept ($p_{\text{BH}} = 0.017$)	a column where fewer than 0.5% of patients differ carries almost no information; the test keeps any rare drug that is still related to readmission instead of dropping it blindly.

Issue found	What we did	Why, and not the alternative
Hidden missing codes ("Not Available", "NULL", "Not Mapped") in 3 ID columns	recoded to missing, together with `"?"`	they are missing data posing as categories; left in, they would look like real admission types. 214,992 raw cells (4.2%) are missing once these are counted.
Missing categorical values	under 5% missing: mode (`race`, `gender`, `diag_1-3`, discharge); 5% or more: an explicit `Unknown` level (`medical_specialty` 48%, `payer_code` 43%, `admission_type` 11%, `admission_source` 7%)	the mode cannot distort a column that is 95% observed, but imputing 48% would invent a specialty for half the patients. Missingness is informative: early readmission is 9.9% when `payer_code` is missing and 8.3% when recorded.
Out-of-range values?	range checks on all 8 counts; none found	stays 1–14 days and non-negative counts are all valid.
Multivariate outliers in the 8 counts	standardised, projected to 2 principal components (41% of variance), Isolation Forest with 1% contamination: 700 patients removed	removes extreme combinations that no single-column rule sees. The 700 are mostly heavy users (mean 2.1 prior admissions and 1.9 emergency visits, vs 0.16 and 0.09); see Section 6 for what this costs.
Text categories for modelling	`age` ordinal 0–9; `change`, `diabetesMed` 0/1; 18 nominal columns one-hot (110 columns); `medical_specialty` and `diag_1-3` integer indices (71 / 694 / 722 / 756 levels); counts capped at P99.5 in the model matrix	one-hot on the four high-cardinality columns would add 2,243 sparse columns, so they are indexed for an embedding layer. The caps only affect the model matrix; the tests use the uncapped cleaned table.

3. What changed after preparation

Measure	Before	After
Unit of a row	encounter	patient (first encounter)
Rows	101,766	69,290
Encounters per patient (max)	40	1
Columns	50	35 cleaned; 128 encoded
Missing cells	214,992 (4.2%)	0; 76,206 cells (3.1%) in an explicit `Unknown`
Early readmission (`<30`)	11.16%	8.86% (6,142 patients)
Mean `number_inpatient`	0.64 (SD 1.26)	0.16 (SD 0.51)
No prior inpatient admission	66.5%	88.7%
Mean `number_emergency`	0.20	0.09
Mean length of stay	4.40 days	4.25 days
Mean lab procedures	43.1	42.7

Moving from encounters to patients changed the outcome the most: the early-readmission rate fell from 11.2% to 8.9% (Figure 1). The raw rate is inflated because patients who are admitted again and again contribute many rows and are readmitted more often. The same effect, plus the outlier screen, cut the mean number of prior admissions by three quarters. Stay-level variables barely moved (length of stay 4.40→4.25 days), so cleaning did not distort them.

4. Basic questions and problem statement

We first asked seven descriptive questions, one per plot (Figures 1–7), then used what they showed to frame three problem statements that a test could answer either way.

Basic questions

1. What share of patients are readmitted, and within 30 days? (Fig. 1)
2. How does readmission vary with age? (Fig. 2)
3. Do readmitted patients stay longer in hospital? (Fig. 3)
4. Do readmitted patients use more care in the year before? (Fig. 4)
5. Is medication management related to readmission? (Fig. 5)
6. Do HbA1c and glucose results separate the outcomes? (Fig. 6)
7. Which numeric features move together, and which tracks early readmission? (Fig. 7)

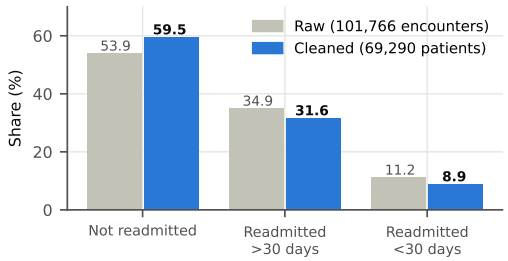


Figure 1: Q1. About 1 patient in 11 (8.9%) is readmitted within 30 days; counting patients instead of encounters lowers that rate from 11.2%, and the 9 : 91 split means very unequal groups in Section 5.

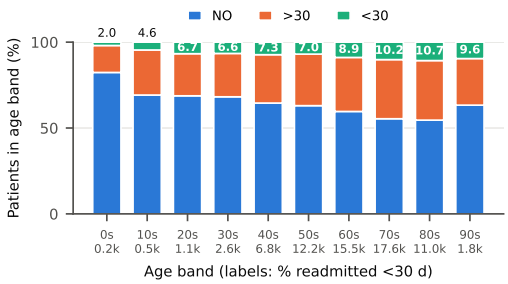


Figure 2: Q2. Early readmission rises with age, from 2.0% in children to 10.7% at 80–89, and the share never readmitted falls from 82% to 55%: age is a likely confounder of every comparison below.

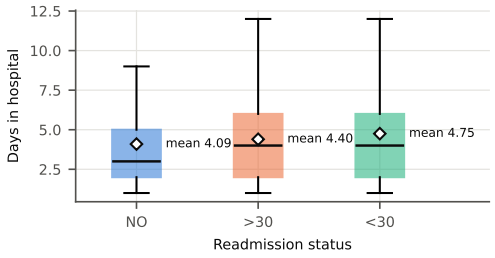


Figure 3: Q3. Median stay is 3 days for NO and 4 days for both readmitted groups; means rise 4.09→4.40→4.75, but the boxes overlap heavily (SD about 3 days in each).

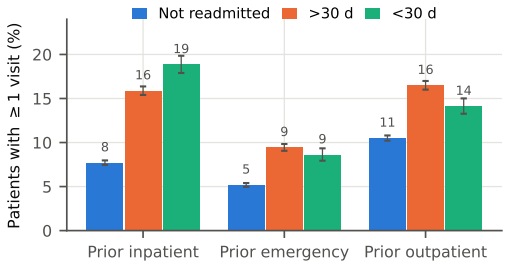


Figure 4: Q4. Only prior *inpatient* admissions rise in step with the outcome (7.7%→15.9%→18.9%); emergency and outpatient visits do not separate <30 from >30. Error bars: 95% Wilson CI.

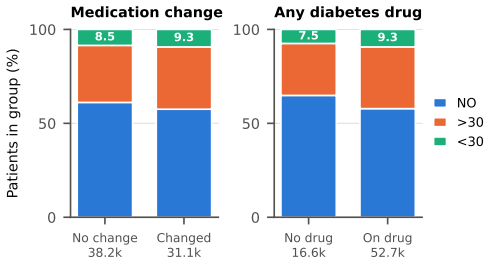


Figure 5: Q5. Early readmission (green labels, %) is 9.3% vs 8.5% with and without a medication change, and 9.3% vs 7.5% with and without any diabetes drug: differences of only 1–2 points.

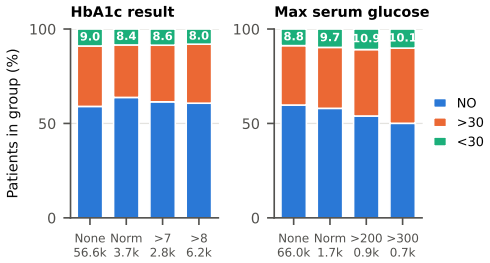


Figure 6: Q6. HbA1c was not measured for 82% of patients and glucose for 95%; early readmission is 8.0–9.0% across HbA1c results, so neither lab result is a useful test variable.

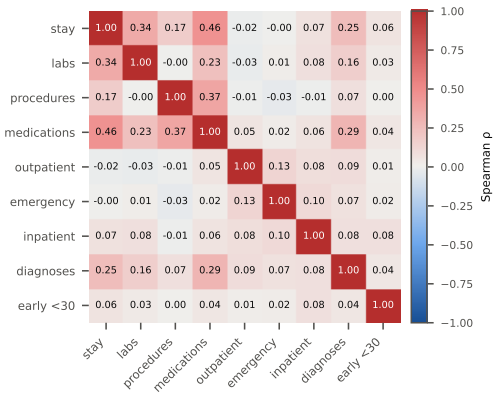


Figure 7: Q7. Stay, medications, labs and procedures move together (ρ up to 0.46); every correlation with early readmission is small, the largest being prior inpatient admissions ($\rho = 0.08$) and stay (0.06).

Our problem statements

P1 (primary). Is an inpatient admission in the previous year a marker of readmission within 30 days, large enough to be useful for flagging patients at discharge, and does risk climb with the number of prior admissions?

P2. Do patients readmitted within 30 days use more hospital resources in their index stay (a stay at least one day longer, more variable lab testing), and do the resource measures move together?

P3. Is how a patient's diabetes is being managed (insulin dose changes; any diabetes drug) related to early readmission by a margin that matters?

How they are novel: the basic questions describe one variable at a time. Ours compare the clinically relevant groups (`<30` against `NO`) on the *patient-level* sample, with the smallest effect that would matter fixed in advance, so each could come out "significant but unimportant". P1 targets a feature a discharge nurse can check in seconds, and Figure 4 shows why we chose inpatient rather than emergency or outpatient history.

5. Statistical inferences

Rows tested: all 69,290 patients of the cleaned table, before encoding and capping. Tests comparing two outcome groups use `<30` (6,142) against `NO` (41,239); `>30` (21,909) is left out there because those patients *were* readmitted, just later, which makes them neither a case nor a clean control.

Why the thresholds matter here: with groups this large a difference of a few tenths of a percentage point is "significant". Every decision on importance therefore compares the 95% CI with a threshold fixed before running the test. For association strength we use Cohen's conventional small effect (Cramér's $V = 0.10$).

Test 1 (P1): do prior inpatient admissions follow a Poisson distribution?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	`number_inpatient` $\sim$ Poisson(λ), λ estimated from the sample
$\boldsymbol{H_1}$	its distribution differs from that Poisson
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2 (any departure)
Test	Pearson χ^2 goodness of fit on 0 / 1 / 2+
Smallest effect that matters	Cohen's $w = 0.10$

This test decides how P1 should treat the count. If it is Poisson, comparing mean counts is fine; if not, Tests 2 and 3 should use categories. Values of 2 and above are pooled so every expected count is above 5; d.f. $= 3 - 1 - 1 = 1$ because λ is estimated.

b) Other tests we considered

A discrete variable checked against a fitted discrete distribution, with large expected counts (smallest 768): a χ^2 goodness-of-fit test fits.

Also considered	Why we did not use it
χ^2 test for variance	checks only one Poisson property (variance $= \lambda$), not the shape; run anyway, $\chi^2 = 114{,}636$ (69,289), $p < 10^{-300}$, variance is 1.65 times the mean
One-sample t-test	tests only whether the mean equals a value; a Poisson with the right mean can still have the wrong shape
z-test	same limitation as the t-test: about the mean only
χ^2 test for independence	needs two variables; here one variable is checked against a model

c) Result

Estimate	$w = 0.233$; $\hat{\lambda} = 0.157$ (95% CI 0.153–0.161); observed 61,475 / 5,719 / 2,096 vs expected 59,231 / 9,291 / 768
95% confidence interval	w: 0.225 to 0.241 (bootstrap, λ refitted each time)
Test statistic (d.f.)	$\chi^2 = 3{,}752$ (1)
p-value	$< 10^{-300}$ (below double precision)
Decision	reject H_0

The count is not Poisson, and the departure is large (whole interval above 0.10). There are too many zeros, too few single admissions and 2.7 times too many patients with two or more (variance/mean $= 1.65$). A mean count would be driven by a few heavy users, so Tests 2–3 use categories.

Test 2 (P1): are prior inpatient admissions more common among early readmissions?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_{<30} = p_{NO}$
$\boldsymbol{H_1}$	$p_{<30} > p_{NO}$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	pooled two-proportion z-test
Smallest difference that matters	5 percentage points

p: share of patients with ≥ 1 inpatient admission in the previous year. One-sided, because P1 claims prior admissions are a *risk* marker. 5 points, because a flag only a few points more common in cases than in controls would mostly mark patients who never return.

b) Other tests we considered

Two independent groups, a binary outcome and large counts (every cell $\geq$ 1,158): a two-proportion z-test, the test that directly encodes a directional H_1.

Also considered	Why we did not use it
χ^2 test for independence (2×2)	gives $\chi^2 = z^2$, the same evidence, but only two-sided; it cannot express "greater than"
Unpaired t-test on mean counts	Test 1 shows the count is over-dispersed, so a mean is driven by a few heavy users
Paired t-test	nothing to pair: each patient is in one group
χ^2 goodness of fit	compares one sample with a fixed distribution, not two groups with each other

c) Result

Estimate	$+11.1$ points (18.9% = 1,158/6,142 vs 7.7% = 3,178/41,239)
95% confidence interval	$+10.2$ to $+12.2$ points (Newcombe)
Test statistic (d.f.)	$z = 28.27$ (no d.f.: normal reference)
p-value	4.4×10^{-176}
Decision	reject H_0

Early-readmitted patients are 2.4 times as likely to have had a prior admission. This matters: the whole interval lies above 5 points, more than twice the threshold.

Test 3 (P1): does readmission depend on how many prior admissions a patient had?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	prior admissions (0 / 1 / 2+) and `readmitted` (`NO`/`>30`/`<30`) are independent
$\boldsymbol{H_1}$	they are associated
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2 (any departure)
Test	Pearson χ^2 test of independence, 3×3
Smallest effect that matters	Cramér's $V = 0.10$

This uses all three outcomes, so it also shows where `>30` patients sit.

b) Other tests we considered

Two nominal variables with large expected counts (minimum 185.8): Pearson χ^2, sized with Cramér's V.

Also considered	Why we did not use it
χ^2 goodness of fit	needs a fixed expected distribution; the question is whether two variables are related
Pairwise z-tests	nine comparisons where one test suffices; more false positives
Unpaired t-test	compares two means; here both variables are categories and there are three outcome groups

c) Result

Estimate	$V = 0.104$; `<30` rate 8.1% (0 prior), 12.8% (1), 20.4% (2+)
95% confidence interval	V: 0.099 to 0.110 (row bootstrap, 500 resamples)
Test statistic (d.f.)	$\chi^2 = 1{,}508.8$ (4)
p-value	$< 10^{-300}$
Decision	reject H_0

A clear dose–response: two or more prior admissions give $2.5\times$ the rate of none (standardised residual $+17.7$ in the 2+/`<30` cell). The point estimate just passes $V = 0.10$ but the interval reaches 0.099: *small, borderline meaningful*.

Test 4 (P2): are early-readmitted patients' stays at least a day longer?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_{<30} = \mu_{\text{NO}}$ (mean length of stay)
$\boldsymbol{H_1}$	$\mu_{<30} \neq \mu_{\text{NO}}$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	Welch two-sample t-test
Smallest difference that matters	1 day

Two-sided, because a short stay (early discharge) is as plausible a cause of readmission as a long one. One day, because stays are recorded in whole days and a bed-day is what hospitals plan for.

b) Other tests we considered

Two independent groups, a numeric response and unknown σ: a two-sample t-test. With $n = 6{,}142$ and 41,239 the means are near normal despite the skew.

Also considered	Why we did not use it
z-test	needs known population σ; at this size it gives the same answer as t
Pooled (equal-variance) t-test	an F-test rejects equal variances ($F = 1.11$, $p = 2.6 \times 10^{-8}$); run anyway, $t = 16.75$, same decision
Paired t-test	no patient is in both groups
One-sample t-test	there is no fixed reference stay to compare against

c) Result

Estimate	$+0.66$ day (4.75 vs 4.09; medians 4 vs 3)
95% confidence interval	$+0.58$ to $+0.74$ day
Test statistic (d.f.)	$t = 16.10$ (7,875)
p-value	2.1×10^{-57}
Decision	reject H_0

Early-readmitted patients stay longer, but not meaningfully so: the whole interval lies below 1 day (Cohen's $d = 0.22$).

Test 5 (P2): is lab testing more variable for early-readmitted patients?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\sigma^2_{<30} = \sigma^2_{\text{NO}}$ (variance of `num_lab_procedures`)
$\boldsymbol{H_1}$	$\sigma^2_{<30} \neq \sigma^2_{\text{NO}}$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	two-sample F-test for equal variances
Smallest difference that matters	variance ratio below 0.80 or above 1.25

If early-readmitted patients are a more mixed group, their lab workload should be more spread out, not just higher. A 25% change in variance (SD about 12%) is our smallest notable change.

b) Other tests we considered

Two independent samples and a question about spread: the F-test compares the two sample variances directly. Its normality assumption is roughly met here: the lab counts are bell-shaped with little skew (-0.28 and -0.19).

Also considered	Why we did not use it
χ^2 test for variance	compares one sample's variance with a known value; we compare two samples with each other
Unpaired t-test	compares means, not spreads; means differ by 2.8 labs (44.7 vs 41.9)
z-test	also about means, and needs known σ
Paired t-test	no pairing between the groups

c) Result

Estimate	variance ratio 0.94 (369.5 vs 392.5; SD 19.2 vs 19.8)
95% confidence interval	0.907 to 0.978 (from the F distribution)
Test statistic (d.f.)	$F = 0.942$ (6,141, 41,238)
p-value	0.0020
Decision	reject H_0

The spreads differ, but in the opposite direction to the idea and by only 6%: the whole interval lies inside 0.80–1.25, so the difference does not matter.

Test 6 (P2): do longer stays go with more medications?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\rho_s = 0$ between `time_in_hospital` and `num_medications`
$\boldsymbol{H_1}$	$\rho_s \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	t-test for the correlation coefficient, on Spearman's rank ρ_s
Smallest effect that matters	$\lvert\rho_s\rvert = 0.10$

P2 treats stay and drug count as two measures of resource use. If they move together, a longer stay is not a separate signal from a heavier drug load.

b) Other tests we considered

Two skewed counts with many ties: the t-test for a correlation coefficient, $t = \rho_s\sqrt{(n-2)/(1-\rho_s^2)}$, applied to ranks so it does not assume a straight line.

Also considered	Why we did not use it
t-test on Pearson's r	measures only a straight-line link and is pulled by outliers in skewed counts; run anyway, $r = 0.454$, same decision
χ^2 test for independence	needs binning both variables, which throws away their order
Paired t-test	would compare mean days with mean drug counts, which are in different units

c) Result

Estimate	$\rho_s = 0.458$ ($n = 69{,}290$)
95% confidence interval	0.452 to 0.464 (Fisher z)
Test statistic (d.f.)	$t = 135.6$ (69,288)
p-value	$< 10^{-300}$
Decision	reject H_0

A moderate, clearly meaningful link: longer stays come with more drugs. Together with Test 4, this says stay length mostly tracks how sick a patient is.

Test 7 (P3): is a change in diabetes medication associated with readmission?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	`change` (`Ch`/`No`) and `readmitted` are independent
$\boldsymbol{H_1}$	they are associated
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2 (any departure)
Test	Pearson χ^2 test of independence, 2×3
Smallest effect that matters	Cramér's $V = 0.10$

b) Other tests we considered

A binary and a three-level variable with large expected counts (minimum 2,753): Pearson χ^2.

Also considered	Why we did not use it
z-test for two proportions	uses only `<30` vs not, throwing away the `>30` class
χ^2 goodness of fit	needs a fixed expected distribution, not a relation between two variables
Unpaired t-test	the outcome is a category, so there is no mean to compare

c) Result

Estimate	$V = 0.037$; `<30` rate 9.3% with a change vs 8.5% without
95% confidence interval	V: 0.030 to 0.043 (row bootstrap)
Test statistic (d.f.)	$\chi^2 = 92.9$ (2)
p-value	6.6×10^{-21}
Decision	reject H_0

A real but negligible association: the whole interval is below half the threshold.

Test 8 (P3): is the insulin dosing category associated with readmission?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	`insulin` (`No`/`Steady`/`Up`/`Down`) and `readmitted` are independent
$\boldsymbol{H_1}$	they are associated
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2 (any departure)
Test	Pearson χ^2 test of independence, 4×3
Smallest effect that matters	Cramér's $V = 0.10$

b) Other tests we considered

Two nominal variables with large expected counts (minimum 587): Pearson χ^2 with Cramér's V.

Also considered	Why we did not use it
Pairwise z-tests	six comparisons where one test suffices; more false positives
χ^2 goodness of fit	needs a fixed expected distribution, not a relation between two variables
Unpaired t-test	insulin categories have no numeric scale

c) Result

Estimate	$V = 0.033$; `<30` rate 8.2% (No), 9.1% (Steady), 9.6% (Up), 10.4% (Down)
95% confidence interval	V: 0.028 to 0.038 (row bootstrap)
Test statistic (d.f.)	$\chi^2 = 146.7$ (6)
p-value	3.8×10^{-29}
Decision	reject H_0

Associated, but very weakly (the whole interval is under two fifths of the threshold). A lowered dose has the highest rate, 2.2 points above no insulin: a marker of unstable disease, not a lever.

Test 9 (P3): are patients on diabetes medication readmitted early more often?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_{\text{Yes}} = p_{\text{No}}$ (share readmitted `<30`)
$\boldsymbol{H_1}$	$p_{\text{Yes}} \neq p_{\text{No}}$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	pooled two-proportion z-test
Smallest difference that matters	2 percentage points

`NO` and `>30` both count as "not early". Two-sided, because medication could mark sicker patients or better-managed ones. 2 points is about a fifth of the 8.9% base rate.

b) Other tests we considered

Two independent groups, a binary outcome and large counts: a two-proportion z-test with a Newcombe interval, which stays accurate when proportions are near 0.

Also considered	Why we did not use it
χ^2 test for independence (2×2)	same two-sided result ($\chi^2 = z^2$) without a signed difference and CI
Unpaired t-test on 0/1 outcome	approximates the z-test ($t = 7.69$, same decision), but the z-test is the standard test for proportions
Paired t-test	no patient is in both groups

c) Result

Estimate	$+1.85$ points (9.31% = 4,904/52,697 vs 7.46% = 1,238/16,593)
95% confidence interval	$+1.37$ to $+2.31$ points (Newcombe)
Test statistic (d.f.)	$z = 7.29$ (no d.f.)
p-value	3.0×10^{-13}
Decision	reject H_0

Higher on medication, but whether it matters is *undecided*: the estimate is just under 2 points and the interval (1.4–2.3) spans the threshold.

Summary of the nine tests

Benjamini–Hochberg adjustment over all nine leaves every decision unchanged (largest adjusted $p = 0.0020$).

Test	p_{BH}	Estimate [95% CI]; matters?
1 Poisson fit of prior admissions	$< 10^{-300}$	$w = 0.23$ [0.22, 0.24]; yes, not Poisson
2 prior admission, `<30` vs `NO`	10^{-175}	$+11.1$ pp [10.2, 12.2]; yes
3 prior admissions × outcome	$< 10^{-300}$	$V = 0.104$ [0.099, 0.110]; borderline
4 length of stay	4×10^{-57}	$+0.66$ d [0.58, 0.74]; no
5 spread of lab tests	0.0020	ratio 0.94 [0.91, 0.98]; no
6 stay vs medications	$< 10^{-300}$	$\rho_s = 0.46$ [0.45, 0.46]; yes
7 medication change	8×10^{-21}	$V = 0.037$ [0.030, 0.043]; no
8 insulin category	6×10^{-29}	$V = 0.033$ [0.028, 0.038]; no
9 any diabetes drug	3×10^{-13}	$+1.85$ pp [1.37, 2.31]; undecided

What the tests say together

All nine nulls are rejected, which is expected with about 69,000 patients. Judged against the thresholds, only prior inpatient use clearly separates early readmissions (Tests 2–3). Stay length, lab testing and medication management differ by amounts too small to matter, and Test 6 suggests they mostly reflect how sick a patient is.

6. Limitations

- **Association, not cause.** Prior admissions, insulin dose changes and medication all track how sick a patient is. Nothing here shows that changing them would change readmission.
- **No adjustment yet.** Age rises with every risk factor (Fig. 2) and all tests are unadjusted. A logistic model with age, diagnoses and discharge type is the natural next step.
- **The outlier screen removed the heaviest users.** The 700 patients dropped by the Isolation Forest averaged 2.1 prior admissions. Removing them most likely *shrinks* the P1 effect, so Tests 2–3 are conservative, but our results do not describe the extreme frequent attenders.
- **First encounters only.** Conclusions apply to a patient's first recorded encounter, not to repeat visits, where readmission is more common (11.2% raw).
- **Imputation.** Mode imputation (e.g. `race`, 2.7% missing) slightly over-counts the majority level; `Unknown` levels are kept but not studied.
- **Large-sample significance.** With $n \approx 47{,}000$ per comparison every test rejects; our importance thresholds are our own conventions, and a different threshold could change the "matters?" column for Tests 3 and 9.
- **Old, US-only data.** The records end in 2008 and come from US hospitals; practice has changed since and may differ elsewhere.
- **Encoding gap.** `admission_type` is in the cleaned table but missing from the encoded matrix; it will be added before modelling.